

MARYAM SAADAT CHEEMA

• mcheema2@asu.edu • linkedin.com/in/maryam-saadat-cheema/ • maryamsaadat.github.io/

RESEARCH INTERESTS

Human–AI Interaction, Accessibility, Evaluation of Generative AI Systems

EDUCATION

Ph.D. in Computer Science, Advisors: Dr. Hasti Seifi & Dr. Pooyan Fazli Jan 2025 - May 2028 (Expected)

Arizona State University, Tempe, AZ

M.S. Computer Science, Advisor: Dr. Hasti Seifi

Aug 2022 - Dec 2024

Arizona State University, Tempe, AZ

B.S. Computer Science

Aug 2018 - May 2022

Lahore University of Management Sciences, Lahore, Pakistan

CONFERENCE PUBLICATIONS

- DescribePro: Collaborative Audio Description with Human–AI Interaction. *ASSETS 2025*. (Maryam S. Cheema, Sina Elihanamesh, Samuel Martin, Pooyan Fazli, Hasti Seifi).
- DescribeNow: User-Driven Audio Description for Blind and Low Vision Individuals. *DIS 2025*. (Maryam S. Cheema, Hasti Seifi, Pooyan Fazli).
- Videoa11y: Method and Dataset for Accessible Video Description. *CHI 2025*. (Chaoyu Li, Sid Padmanabhuni, Maryam S. Cheema, Hasti Seifi, Pooyan Fazli).

RESEARCH EXPERIENCE

Arizona State University, Tempe, AZ: Graduate Researcher, HCI

Mar 2023 – Present

- Developed ViDScribe, a platform to empower blind and low vision users by providing AI-generated audio descriptions.
- Designed and tested prototype for customizable audio description (AD), allowing BLV users to select concise or detailed AD during video playback.
- Designed and tested prototype to test collaborative AD authoring system that enables describers to iteratively refine AI-generated descriptions through multimodal large language model (LLM) prompting and manual editing.
- Developed audio description guidelines for prompting LLMs, used in VideoA11y-40K, the first large-scale video description models training dataset of 40,000 videos.

WORK EXPERIENCE

AI Cloud Innovation Center, Tempe, AZ: UI/UX Designer & Developer

Aug 2023 – Nov 2024

- Built reusable ReactJS components for generative AI chat platforms. Led development of 3 GenAI apps for government, and enterprise clients using AWS Services.
- Designed and prototyped six smart city projects using Figma.

TEACHING

Arizona State University, Tempe, AZ: Teaching Assistant – HCI

- Teaching assistant for Human-Computer interaction for Fall 2023, Spring 2024, Spring 2025, Fall 2025

TECHNICAL SKILLS

Programming: Javascript, C++, Python, HTML, CSS

Design Tools: Figma, Adobe XD, Canva